Write your name here

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# Mathematics

## Level 2



30 April – 4 May 2018
**Time: 1 hour 30 minutes**

Paper Reference
**FSM02/01**

**You must have:**
Pen, calculator, HB pencil, eraser, ruler graduated in cm and mm, protractor, compasses.

Total Marks
48 /48

**My signature confirms that I will not discuss the content of the test with anyone until the end of the 5 day test window.**

**Signature:** Master Functional Skills (MFS)

## Instructions

- Use a **black** ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Sign the declaration.
- Answer **all** questions.
- Answer the questions in the spaces provided
  *– there may be more space than you need.*
- **Calculators may be used.**

## Information

- The total mark for this paper is 48.
- The marks for each question are shown in brackets
  *– use this as a guide to how much time to spend on each question.*
- **You must show clearly how you get your answers because marks will be awarded for your working out.**
- **Check your working and your answers at each stage.**
- **This sign shows where marks will be awarded for showing your check.**

## Advice

- Read each question carefully before you start to answer it.
- Keep an eye on the time.

*Turn over* ▶

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## SECTION A: Car trader

**Answer all questions in this section.**

**Write your answers in the spaces provided.**

**1** Adam works for a car trading company.

He has this information about the number of cars sold by the company in the first six months of 2017

| | Jan | Feb | Mar | Apr | May | Jun |
|---|---|---|---|---|---|---|
| **Number of cars sold** | 31 | 28 | 46 | 52 | 44 | 62 |

Adam says

> The mean number of cars sold each month in this period was more than 42
>
> The number of cars sold in June 2017 was more than $\frac{1}{5}$ of the number of cars sold from the beginning of January to the end of June.

(a) Is Adam correct?
Show why you think this. **(4)**

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Use the box below to show clearly how you get your answer.

*Part 1: Check Statement 1 – Mean monthly cars sold > 42* *[M1 sum]*

**Total cars sold (Jan–Jun) = 31 + 28 + 46 + 52 + 44 + 62**

**Total cars sold = 263 cars**

*Calculate mean monthly sales (Total ÷ 6 months):* *[M1 mean]*

**Mean = 263 ÷ 6 = 43.833... ≈ 43.8 cars per month**

**43.83 > 42 (Mean is greater than 42 by 1.83 cars) Statement 1 is TRUE**

*Part 2: Check Statement 2 – June sales > 1/5 of total 6-month sales* *[M1 fraction]*

**Number of cars sold in June = 62 cars**

**1/5 of total sales = 263 ÷ 5 = 52.6 cars**

**62 > 52.6 (June sales exceeded 1/5 by 9.4 cars) Statement 2 is TRUE**

*Alternative percentage check for Statement 2:*

**June proportion = (62 ÷ 263) × 100 = 23.57...% > 20% (1/5)**

*Final Conclusion & Decision:*

*[A1 decision]*

**Conclusion: Yes, Adam is correct on both statements.**

**(Mean was 43.83 > 42, and June sales of 62 were > 52.6).**

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Adam has an annual target to sell 94 cars a year.

Adam gets a bonus if he meets his target.
His bonus is

- 1.75% of the total sales he makes in a year
- an extra £150 for each car he sells over his target.

In 2017 Adam sold 114 cars and the total sales he made were £316 400

Adam thinks he will get a bonus of more than £8000 for 2017

(b) Will Adam get a bonus of more than £8000?

**(4)**

Use the box below to show clearly how you get your answer.

*Step 1: Calculate percentage bonus (1.75% of total annual sales)* *[M1 sales %]*

**Sales Bonus = 1.75% of £316,400**

**= 0.0175 × £316,400 = £5,537.00**

*Step 2: Calculate extra car bonus (£150 per car sold over target of 94)* *[M1 extra cars]*

**Cars over target = 114 − 94 = 20 extra cars**

**Extra Car Bonus = 20 × £150 = £3,000.00**

*Step 3: Calculate total bonus for 2017* *[M1 total bonus]*

**Total Bonus = £5,537 + £3,000 = £8,537.00**

*Step 4: Compare total bonus with Adam's claim (> £8,000)*

**£8,537 > £8,000 (exceeds £8,000 target by £537.00)**

*[A1 decision]*

**Conclusion: Yes, Adam will get a bonus of more than £8,000.**

*Verification: £8,000 − £5,537 = £2,463 needed; £2,463 ÷ £150 = 16.42 cars; 20 > 16.42 cars (verified)*

**(Total for Question 1 is 8 marks)**

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**2** Adam offers 3 payment options to car buyers.

The payment options are

A) cash
B) instalments
C) lease.

Adam sold 114 cars last year

- 19 cars with option A
- 57 cars with option B
- the remaining cars with option C.

Adam is going to write a report for his manager.
He needs to show the ratio of the number of cars sold using each payment option in the form A : B : C

Write down the ratio in the form A : B : C
Give the ratio in its simplest form. **(3)**

Use the box below to show clearly how you get your answer.

*Step 1: Calculate the number of cars sold with option C (lease)* *[M1 option C]*

**Total cars sold = 114 cars**

**Option A (cash) = 19 cars, Option B (instalments) = 57 cars**

**Option C = 114 − 19 − 57 = 38 cars**

*Step 2: Write down the unsimplified ratio A : B : C* *[M1 ratio]*

**Ratio A : B : C = 19 : 57 : 38**

*Step 3: Simplify the ratio by dividing by highest common factor (19)*

**19 ÷ 19 = 1, 57 ÷ 19 = 3, 38 ÷ 19 = 2**

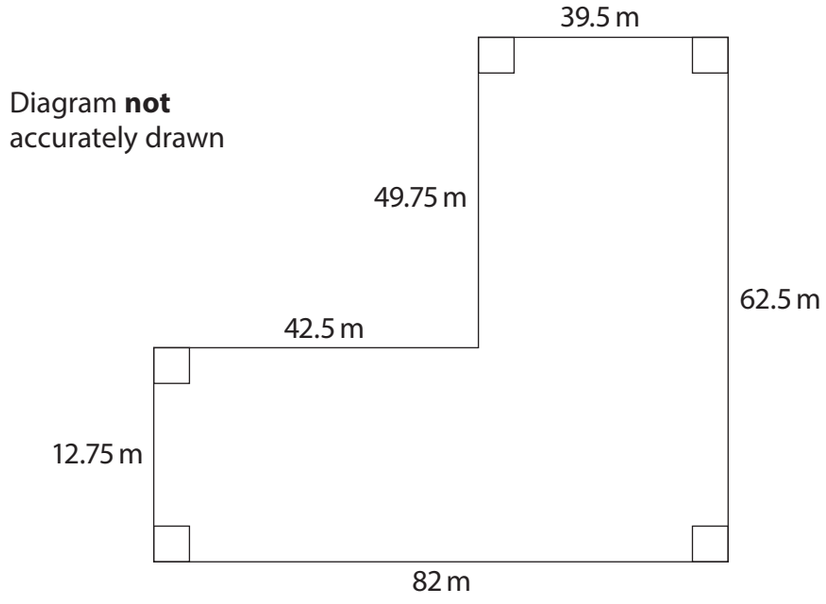
**Simplest Form Ratio = 1 : 3 : 2** *[A1 answer]*

*Check: Total parts = 1 + 3 + 2 = 6 parts; 114 ÷ 6 = 19 cars per part (19, 57, 38) (verified)*

**(Total for Question 2 is 3 marks)**

**3** Kali works for a building company.
She needs to put a new surface on the car park of the car trading company.

The diagram shows the section of the car park that needs a new surface.




Kali will cover the whole section of the car park with concrete.
The concrete will be 0.09 m deep.

What is the total volume of concrete Kali will need?
Show a check of your working. **(5)**

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Use the box below to show clearly how you get your answer.

*Step 1: Split L-shaped car park into two simple rectangles* *[M1 split]*

**Rectangle 1 (Left): 39.5 m wide by 62.5 m long**

**Area 1 = 39.5 × 62.5 = 2,468.75 m²**

**Rectangle 2 (Bottom right): 42.5 m wide by 12.75 m long**

**Area 2 = 42.5 × 12.75 = 541.875 m²**

*Step 2: Calculate total surface area of car park* *[M1 total area]*

**Total Area = 2,468.75 + 541.875 = 3,010.625 m²**

**(Alternative: Full bounding box 82 × 62.5 − cutout 42.5 × 49.75 = 5,125 − 2,114.375 = 3,010.625 m²)**

*Step 3: Calculate volume of concrete (Area × Depth)* *[M1 volume]*

**Depth of concrete = 0.09 m**

**Volume = 3,010.625 m² × 0.09 m**

**= 270.95625 m³**

*Final Answer with appropriate rounding & units:*

**Total Volume of Concrete = 270.96 m³ (or 271 m³)** *[A1 answer]*

Use the box below to show your check.

☑

*Reverse division check:* *[B1 check]*

**270.95625 ÷ 0.09 = 3,010.625 m² (matches area exactly)**

**Estimation: 3,000 m² × 0.1 m = 300 m³ ≈ 271 m³ (realistic) (correct)**

**(Total for Question 3 is 5 marks)**

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**SECTION B: Theatre**

**Answer all questions in this section.**

**Write your answers in the spaces provided.**

**4** Catherine works in a theatre.
She sets the price of tickets for a new show.

Catherine uses this formula to find the price of a ticket for a seat in the stalls.

$$P = \frac{6500}{x + 0.9y}$$

P is the normal price of a ticket for a seat in the stalls (£)
$x$ is the total number of seats in the stalls
$y$ is the total number of seats in the balcony

The theatre has

- 180 seats in the stalls
- 70 seats in the balcony.

On Monday Catherine sells 45 tickets for seats in the stalls to a school.
She charges 60% of the normal ticket price for each seat.

(a) Work out the total amount Catherine should charge the school.

**(5)**

Use the box below to show clearly how you get your answer.

*Step 1: Substitute seats into formula denominator: x + 0.9y* *[M1 sub]*

**x = 180 stalls seats, y = 70 balcony seats**

**Denominator = 180 + (0.9 × 70) = 180 + 63 = 243**

*Step 2: Calculate normal ticket price for stalls seat (P)* *[M1 normal P]*

**P = 6,500 ÷ 243 = £26.74897... per ticket**

*Step 3: Apply 40% discount for school (school pays 60% of normal price)* *[M1 discount]*

**Discounted Price = 0.60 × £26.74897... = £16.04938... per ticket**

**Total for 45 tickets = 45 × £16.04938... = £722.22**
*[A1 answer]*

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Catherine checks the formula.
She realises the figure 6500 should have been 6000

(b) What effect does this error have on the price of a ticket for a seat in the stalls?
**(1)**

Use the box below to write your comment.

*Comment on Effect of Error:* *[A1 comment]*

**The price of a ticket will decrease (it will be lower).**

**Because the numerator is smaller (6000 < 6500), so 6000 ÷ 243 = £24.69 < £26.75.**

Catherine wants to record information about the people who see the new show on Friday.

She wants to know their

- gender (female, male)
- age (16–24, 25 and over)
- rating for the show (excellent, good, poor).

(c) Design a data collection sheet for Catherine.

**(3)**

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Use the box below to show your data collection sheet.

*Two-Way Audience Feedback Collection Sheet (Tally & Frequency):* *[M1 design]*

| | Male | | Female | |
|---|---|---|---|---|
| Show Rating | 16 · 24 | 25 & over | 16 · 24 | 25 & over |
| Excellent | //// /// | //// //// | //// //// | //// //// // |
| Good | //// / | //// // | //// /// | //// / |
| Poor | // | / | /// | // |
| Total Audience | 16 | 18 | 21 | 20 |

*Key Design Features & Standards Met:* *[A1 efficiency]*

- **Gender Categories:**
  Separate primary columns for Female and Male.
- **Age Group Subdivisions:**
  Distinct sub-columns for 16–24 and 25 & over within each gender.
- **Rating Hierarchy:**
  Dedicated row options for Excellent, Good, and Poor with tally spaces and totals.
- **Data Efficiency:**
  Allows instant tally recording as audience members exit the theatre.

**(Total for Question 4 is 9 marks)**

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**5** Robin is going to design a new food court for the theatre.

He has this sketch of the space for the new food court.

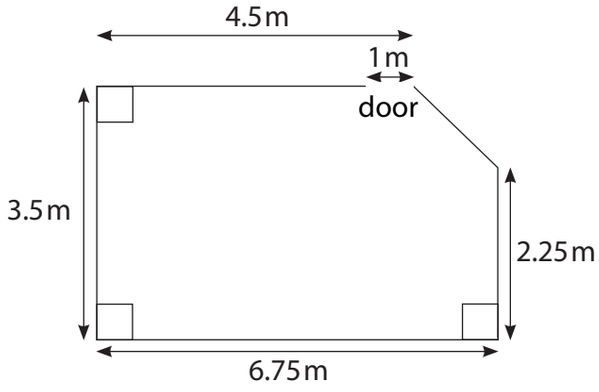



Diagram **not** accurately drawn

Robin starts a scale drawing of this space.

Complete the scale drawing for Robin.
Remember to fill in the key. **(4)**

[A1 complete scale drawing]

Top wall: 4.5 m (9 squares) Door: 1m (2 sq)

Diagonal wall (3.15 sq)

2.25 m (4.5 sq)

Bottom wall: 6.75 m (13.5 squares)

**Key** 1 square length on the page is 0.5 m in the food court in the theatre [B1 key]

3.5 m edge

**(Total for Question 5 is 4 marks)**

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**6** Catherine sells bags of popcorn at the theatre.
The bags come in three sizes.

| small | medium | large |
|---|---|---|
| 60 g | 150 g | 250 g |
| £1.49 | £3.99 | £5.99 |

Catherine thinks the medium bag of popcorn is the best value for money.

Is the medium bag of popcorn the best value for money?

**(3)**

Use the box below to show clearly how you get your answer.

*Step 1: Calculate cost per gram (or price per 100g) for each bag* *[M1 rate]*

- **Large bag (250 g for £5.99):**

  **Cost per gram = £5.99 ÷ 250 g = £0.02396 / g (= £2.40 per 100g)**

- **Medium bag (150 g for £3.99):**

  **Cost per gram = £3.99 ÷ 150 g = £0.02660 / g (= £2.66 per 100g)**

- **Small bag (60 g for £1.49):**

  **Cost per gram = £1.49 ÷ 60 g = £0.02483 / g (= £2.48 per 100g)**

*Step 2: Compare value across all three sizes* *[M1 compare]*

**Lowest cost per gram = Large bag (£0.02396 / g)**

**Highest cost per gram = Medium bag (£0.02660 / g) (Worst Value!)**

*Conclusion & Decision:*

*[A1 decision]*

**Conclusion: No, the medium bag is NOT the best value for money.**

**(The large bag is the best value at 2.40p/g; medium is dearest at 2.66p/g).**

**(Total for Question 6 is 3 marks)**



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## SECTION C: Garden party

**Answer all questions in this section.**

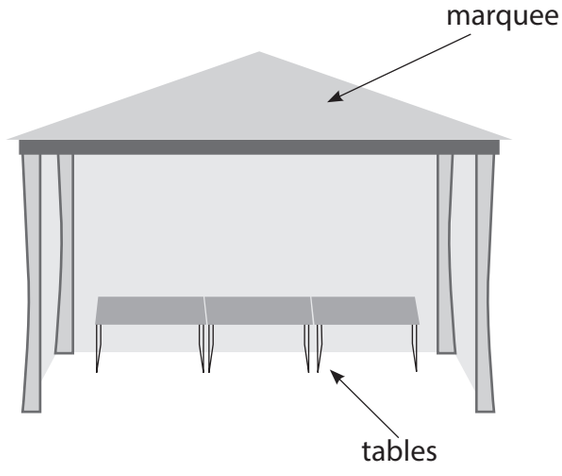
**Write your answers in the spaces provided.**

**7** Fez and Zoe are organising a garden party for their wedding anniversary.

Fez wants to hire a marquee for 2 days.

He finds information about some marquees.

| **marquee** | **A** | **B** | **C** |
|---|---|---|---|
| **size (L by W)** | 6 m by 3 m | 3 m by 3 m | 4 m by 4 m |
| **hire charge per day** | £67 | £59 | £74 |
| **delivery and set up** | £49.95 | £41.99 | £39.99 |




The marquee needs to be big enough to fully cover

- 3 rectangular tables, each 150 cm by 90 cm, joined together along the same edge (short edge or long edge) with no gap between them
- 50 cm free space around the joined tables for people to walk around.

Fez wants to spend the least amount of money for the marquee hire, delivery and set up.

(a) Which marquee should Fez choose?
Show why you think this. **(5)**

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Use the box below to show clearly how you get your answer.

Step 1: Calculate required dimensions for 3 joined tables + 50 cm border [M1 dimensions]

Each table: Length = 150 cm = 1.5 m, Width = 90 cm = 0.9 m

Arrangement 1 (Joined along 90 cm edge in a long line):

- Tables Length = 3 × 1.5 m = 4.5 m, Width = 0.9 m
- Space needed (adding 50 cm = 0.5 m each side):

Required Length = 4.5 + 2 × 0.5 = 5.5 m

Required Width = 0.9 + 2 × 0.5 = 1.9 m --> Fits in 6m × 3m (Marquee A)

Arrangement 2 (Joined along 150 cm edge side-by-side):

- Tables Length = 1.5 m, Width = 3 × 0.9 m = 2.7 m
- Space needed: Length = 1.5 + 1.0 = 2.5 m, Width = 2.7 + 1.0 = 3.7 m

--> Fits in 4m × 4m (Marquee C)

Step 2: Check which marquees can accommodate the tables

- Marquee A (6m × 3m): FITS Arrangement 1 (5.5m ≤ 6m and 1.9m ≤ 3m)
- Marquee B (3m × 3m): TOO SMALL (3m < 5.5m and 3m < 3.7m) --> Eliminated
- Marquee C (4m × 4m): FITS Arrangement 2 (2.5m ≤ 4m and 3.7m ≤ 4m)

Step 3: Compare total hire costs for 2 days + delivery/setup [M1 cost]

- Marquee A: (2 days × £67) + £49.95 = £134.00 + £49.95 = £183.95
- Marquee C: (2 days × £74) + £39.99 = £148.00 + £39.99 = £187.99

Conclusion (Fez wants to spend the least amount of money):

**Fez should choose Marquee A (Total cost £183.95).** [A1 decision]

(Marquee A is £4.04 cheaper than Marquee C: £183.95 < £187.99).

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Fez needs to organise some food for the party.

There will be 80 people at the party.
A catering company has this offer for hot meals

| | |
|---|---|
| non-vegetarian | £19.95 per person |
| vegetarian | £14.95 per person |

Special offer
get $\frac{1}{3}$ off your total bill

Fez uses this offer.

He will order vegetarian meals for 49 people and non-vegetarian meals for the remaining people.

Fez has a budget of £1000 for all the meals.

(b) Is £1000 enough for all the meals?

**(4)**

Use the box below to show clearly how you get your answer.

*Step 1: Calculate normal catering cost for 80 guests* *[M1 meals]*

- 49 Vegetarian meals @ £14.95 = 49 × £14.95 = £732.55
- Non-veg meals = 80 − 49 = 31 guests
- 31 Non-vegetarian meals @ £19.95 = 31 × £19.95 = £618.45

Total Bill before discount = £732.55 + £618.45 = £1,351.00

*Step 2: Apply 1/3 special offer discount (pay 2/3 of bill)* *[M1 discount]*

Discount = 1/3 × £1,351.00 = £450.333...

Total to Pay = £1,351.00 − £450.333... = £900.666... ≈ £900.67

*Step 3: Compare with Fez's budget (£1,000)*

£900.67 < £1,000 (under budget by £1,000 − £900.67 = £99.33)

*[A1 decision]*

Conclusion: Yes, £1,000 is enough for all the meals.

**(Total for Question 7 is 9 marks)**

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**8** Zoe needs to do some things before the party.
This is her list.

- 11:45 am hair appointment at salon (one and a half hours)
- then collect the cake from the bakery (10 minutes)
- finally, buy decorations at the arcade (a quarter of an hour)

She knows that it takes

- 13 minutes to drive from the salon to the bakery
- 12 minutes to drive from the bakery to the arcade
- 16 minutes to drive from the arcade to her home.

These times include time to park her car.

Zoe wants to do all the things on the list and return home by 2:30 pm.

Can Zoe do all the things on the list and return home by 2:30 pm?

**(3)**

Use the box below to show clearly how you get your answer.

*Timeline Schedule (Starting at 11:45 am):* *[M1 time]*

- 11:45 am to 1:15 pm: Salon hair appointment (1 h 30 min = 90 mins)
- 1:15 pm to 1:28 pm: Drive salon to bakery (13 mins)
- 1:28 pm to 1:38 pm: Collect cake at bakery (10 mins)
- 1:38 pm to 1:50 pm: Drive bakery to arcade (12 mins)
- 1:50 pm to 2:05 pm: Buy decorations at arcade (15 mins)
- 2:05 pm to 2:21 pm: Drive arcade to home (16 mins)

*Total Duration & Arrival Calculation:* *[M1 total]*

Total time = 90 + 13 + 10 + 12 + 15 + 16 = 156 mins (2 h 36 min)

Arrival Home = 11:45 am + 2 h 36 min = 2:21 pm (9 minutes before 2:30 pm)

*[A1 decision]*

Conclusion: Yes, Zoe can do all things and return home by 2:30 pm.

**(Total for Question 8 is 3 marks)**

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**9** Zoe prepares some lemonade for the party.

She needs to use $\frac{3}{4}$ of a kilogram of sugar.

Zoe knows that 1 kg is 2.2 pounds.

(a) What is $\frac{3}{4}$ of a kilogram in pounds?
Show a check of your working. **(3)**

Use the box below to show clearly how you get your answer.

*Step 1: Convert fraction 3/4 kg to decimal*

**3/4 kg = 3 ÷ 4 = 0.75 kg**

*Step 2: Multiply by conversion factor (1 kg = 2.2 pounds)* *[M1 convert]*

**Weight in pounds = 0.75 × 2.2**

**= 1.65 pounds**

*Final Answer with units:*

**3/4 kg = 1.65 pounds** *[A1 answer]*

Use the box below to show your check.

☑

*Reverse calculation check:* *[B1 check]*

**1.65 ÷ 2.2 = 0.75 kg = 3/4 kg (matches original quantity exactly)**

**Or: 1/4 kg = 0.55 lb; 3 × 0.55 lb = 1.65 pounds (verified) (correct)**

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Fez has 150 balloons in a bag.
There are 43 red balloons, 67 blue balloons and 40 yellow balloons.

Fez takes one balloon at random from the bag.

(b) What is the probability that the balloon he takes will be blue?

**(1)**

Use the box below to show your answer.

*Probability Calculation: P(Blue Balloon)* *[A1 prob]*

**Number of blue balloons = 67, Total balloons = 150**

**P(Blue) = 67 / 150 (= 0.4467... ≈ 44.7%)**

**(Total for Question 9 is 4 marks)**

**TOTAL FOR PAPER IS 48 MARKS**

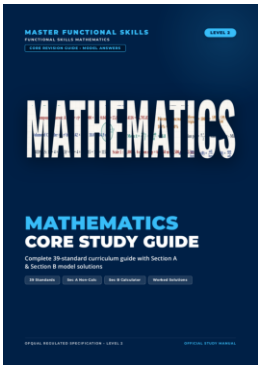
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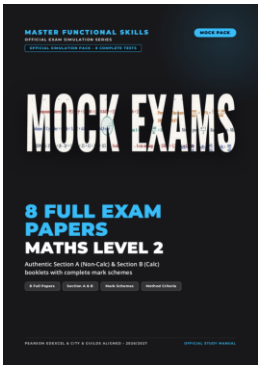



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